

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 213: Digital Electronics

Batch: 25th
Date: 15/03/2025

Final Examination

Semester: Fall 2024
Time: 2 Hours
Total Marks: 40

Answer any five questions:

5 × 8 = 40

- | | CLOs | Marks |
|--|------|-------|
| 1. (a) Illustrate functional diagram of an Arithmetic-Logic Unit. | CLO3 | 4 |
| (b) Design a five-bit parallel adder circuit using full adders. | CLO1 | 4 |
| 2. (a) Sketch a 16-bit synchronous Up/Down counter. | CLO4 | 4 |
| (b) Draw a MOD-25 synchronous Up counter using D flip-flop. | CLO4 | 4 |
| 3. (a) Write the differences between parallel in/parallel out and parallel in/serial out register. | CLO4 | 4 |
| (b) List the commonly used digital IC terminology. | CLO2 | 4 |
| 4. (a) Illustrate a Binary Coded Decimal (BCD) Decoder. | CLO1 | 4 |
| (b) Draw and explain an eight input Encoder. | CLO1 | 4 |
| 5. (a) Write the steps to build a small project management. | CLO1 | 4 |
| (b) List the commonly used IC packages. | CLO2 | 4 |
| 6. (a) Which memory stores the most bits: a 5M × 16 memory or a memory that stores 3M words at a word size of 32 bits? | CLO4 | 4 |
| (b) Draw and explain a Digital-to-Analog Converter. | CLO1 | 4 |
| 7. Write short notes on: | CLO4 | 2×4=8 |
| I. Static RAM | | |
| II. Access time | | |
| III. Auxiliary memory | | |
| IV. Capacity | | |

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 213: Digital Electronics

Batch: 25th
Date: 15/03/2025

Final Examination

Semester: Fall 2024
Time: 2 Hours
Total Marks: 40

Answer any five questions:

5 × 8 = 40

- | | CLOs | Marks |
|--|------|-------|
| 1. (a) Illustrate functional diagram of an Arithmetic-Logic Unit. | CLO3 | 4 |
| (b) Design a five-bit parallel adder circuit using full adders. | CLO1 | 4 |
| 2. (a) Sketch a 16-bit synchronous Up/Down counter. | CLO4 | 4 |
| (b) Draw a MOD-25 synchronous Up counter using D flip-flop. | CLO4 | 4 |
| 3. (a) Write the differences between parallel in/parallel out and parallel in/serial out register. | CLO4 | 4 |
| (b) List the commonly used digital IC terminology. | CLO2 | 4 |
| 4. (a) Illustrate a Binary Coded Decimal (BCD) Decoder. | CLO1 | 4 |
| (b) Draw and explain an eight input Encoder. | CLO1 | 4 |
| 5. (a) Write the steps to build a small project management. | CLO1 | 4 |
| (b) List the commonly used IC packages. | CLO2 | 4 |
| 6. (a) Which memory stores the most bits: a 5M × 16 memory or a memory that stores 3M words at a word size of 32 bits? | CLO4 | 4 |
| (b) Draw and explain a Digital-to-Analog Converter. | CLO1 | 4 |
| 7. Write short notes on: | CLO4 | 2×4=8 |
| I. Static RAM | | |
| II. Access time | | |
| III. Auxiliary memory | | |
| IV. Capacity | | |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science , Engineering and Technology

Bachelor of Science in CSE

BAN211: History of Emergence of Bangladesh

Final Examination

Fall 2025

25th Batch

12 March 2025

3rd Semester

Time: 2 hours

Total Marks: 40

CLOs	Questions	Level	Marks
CLO1	Investigate the causes of the Partition of Bengal (1905) and the reactions it generated among different communities and political groups. Or Identify the key political and historical factors that led to the Lahore Resolution of 1940, and evaluate its significance in shaping the demand for the creation of Pakistan.	Analyze	6
CLO2	Evaluate the significance and achievements of the Language Movement of 1952. How did its outcomes influence the political and cultural identity of Bangladesh? Or Assess the role of the Language Movement of 1952 in the development of Bengali nationalism.	Evaluate	10
CLO3	Examine the Political and Economic disparities imposed by West Pakistan. Or Analyze the exploitation of East Pakistan's resources and the unequal distribution of wealth between East and West Pakistan, and discuss how these factors contributed to the growing demands for autonomy and independence in East Pakistan.	Analyze	8
CLO4	What were the key political, economic, and social factors that triggered the Mass-up surging of 1969 in East Pakistan, and what impact did the movement have on the political landscape of East Pakistan? Or What were the consequences of the movement of Mass-up surging of 1969 for East Pakistan's political situation?	understand	8
CLO4	Explain the key political, economic, and social factors that led to the independence of Bangladesh in 1971. Or How did the disparities between East and West Pakistan contribute to the rise of the nationalist movement and the ultimate separation from Pakistan?	Analyze	8



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science , Engineering and Technology

Bachelor of Science in CSE

BAN211: History of Emergence of Bangladesh

Final Examination

Fall 2025

25th Batch

12 March 2025

3rd Semester

Time: 2 hours

Total Marks: 40

CLOs	Questions	Level	Marks
CLO1	Investigate the causes of the Partition of Bengal (1905) and the reactions it generated among different communities and political groups. Or Identify the key political and historical factors that led to the Lahore Resolution of 1940, and evaluate its significance in shaping the demand for the creation of Pakistan.	Analyze	6
CLO2	Evaluate the significance and achievements of the Language Movement of 1952. How did its outcomes influence the political and cultural identity of Bangladesh? Or Assess the role of the Language Movement of 1952 in the development of Bengali nationalism.	Evaluate	10
CLO3	Examine the Political and Economic disparities imposed by West Pakistan. Or Analyze the exploitation of East Pakistan's resources and the unequal distribution of wealth between East and West Pakistan, and discuss how these factors contributed to the growing demands for autonomy and independence in East Pakistan.	Analyze	8
CLO4	What were the key political, economic, and social factors that triggered the Mass-up surging of 1969 in East Pakistan, and what impact did the movement have on the political landscape of East Pakistan? Or What were the consequences of the movement of Mass-up surging of 1969 for East Pakistan's political situation?	understand	8
CLO4	Explain the key political, economic, and social factors that led to the independence of Bangladesh in 1971. Or How did the disparities between East and West Pakistan contribute to the rise of the nationalist movement and the ultimate separation from Pakistan?	Analyze	8







HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT213

Final Examination

Date: 3/22/2025

Course Title: Statistics and Probability
Semester: Fall 2024
Total Marks: 40

Total Time: 2 Hours

Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- Q1.** (a) The joint probability of x and y is $P(x, y) = C(x + y); x = 1, 2, 3, y = 1, 2$
Find
i. The value of C .
ii. The marginal probability function of x and y .
CLOs CLO3 Marks [5]
- (b) Find harmonic mean, mode and median from the probability density function
 $f(x) = \frac{3}{4}x(2 - x). \quad 0 < x < 2$
CLO3 [5]
- Q2.** (a) If the joint probability function $f(xy) = xe^{-x(y+1)}, x \geq 0, y \geq 0$; find marginal and conditional density functions and also show that x and y are dependent.
CLO2 [5]
- (b) The joint probability function of two random variables x and y is given by
 $f(x, y) = 6xy; 0 \leq x; y \leq 1$
Find $f(x), f(y), f(x/y), f(y/x)$ and check whether x and y are independent variables.
CLO2 CLO3 [5]
- Q3.** (a) Six unbiased coins are tossed simultaneously. What is the probability of getting
i. Exactly three tails.
ii. At least four heads.
CLO2 [3]
- (b) A manufacturer of pins knows that 8% of his product is defective. If he sells pins in boxes of 120 and guarantees that not more than 15 pins will be defective. What is the approximate probability that a box will fail to meet the guaranteed quality?
CLO3 [3]
- (c) The following data show the suicides of 1096 women in 8 cities in a country during 14 years.
CLO3 [4]
- | | | | | | | | | |
|----------------|-----|-----|-----|----|----|----|---|---|
| No of suicides | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Frequency | 364 | 376 | 218 | 89 | 33 | 13 | 2 | 1 |
- Fit a poisson distribution to the above data.
- Q4.** (a) What is independent and dependent events?
CLO1 [2]
- (b) There are two identical boxes containing respectively 5 white and 3 red balls, 4 white and 6 red balls. A box is chosen at random and a ball is drawn at random from it. If the ball is white, what is the probability that it is from the Second box?
CLO3 [3]
- (c) State and prove Bayes theorem.
CLO2 [5]



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT213

Final Examination

Date: 3/22/2025

Course Title: Statistics and Probability

Semester: Fall 2024

Total Marks: 40

Total Time: 2 Hours

$4 \times 10 = 40$

Answer any four (04) out of the following questions:

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

CLOs Marks
CLO3 [5]

Q1. (a) The joint probability of x and y is $P(x, y) = C(x + y); x = 1, 2, 3, y = 1, 2$

Find

- The value of C .
- The marginal probability function of x and y .

CLO3 [5]

(b) Find harmonic mean, mode and median from the probability density function

$$f(x) = \frac{3}{4}x(2-x).$$

$$0 < x < 2$$

Q2. (a) If the joint probability function $f(xy) = xe^{-x(y+1)}, x \geq 0, y \geq 0$; find marginal and conditional density functions and also show that x and y are dependent.

CLO2 [5]

(b) The joint probability function of two random variables x and y is given by $f(x, y) = 6xy; 0 \leq x; y \leq 1$
Find $f(x), f(y), f(x/y), f(y/x)$ and check whether x and y are independent variables.

CLO2 CLO3 [5]

Q3. (a) Six unbiased coins are tossed simultaneously. What is the probability of getting

CLO2 [3]

- Exactly three tails.
- At least four heads.

CLO3 [3]

(b) A manufacturer of pins knows that 8% of his product is defective. If he sells pins in boxes of 120 and guarantees that not more than 15 pins will be defective. What is the approximate probability that a box will fail to meet the guaranteed quality?

CLO3 [4]

(c) The following data show the suicides of 1096 women in 8 cities in a country during 14 years.

No of suicides	0	1	2	3	4	5	6	7
Frequency	364	376	218	89	33	13	2	1

Fit a poisson distribution to the above data.

CLO1 [2]

Q4. (a) What is independent and dependent events?

CLO3 [3]

(b) There are two identical boxes containing respectively 5 white and 3 red balls, 4 white and 6 red balls. A box is chosen at random and a ball is drawn at random from it. If the ball is white, what is the probability that it is from the Second box?

CLO2 [5]

(c) State and prove Bayes theorem.



HAMDARD UNIVERSITY BANGLADESH



Fac
Dep

ch: 25th
Date: 17/03/2025

- Q5. (a) Find the expected value of the number of points that will be obtained in a single toss of a fair dice. CLO3 [2]
- (b) Find the characteristic function of $f(x) = e^{-x}, 0 \leq x \leq \infty$ and hence find mean and variance of $f(x)$ CLO3 [5]
- (b) Two unbiased dice are tossed simultaneously. What is the probability of getting a total of point 10 or even numbers from both the dice? CLO3 [3]

- Q6. (a) Seven coins are tossed at a time and the number of head are noted. The experiment is repeated 130 times and the distribution is obtained as follows: CLO3 [6]

No of heads	0	1	2	3	4	5	6	7
Frequencies	7	6	19	35	30	23	7	1

Fit a binomial distribution to the above data assuming

- The coin is unbiased
 - The nature of the coin is not known.
- (b) A Random variable x has the following probability density function, CLO3 [4]
- $$f(x) = cx(2 - x), 0 \leq x \leq 2$$
- Determine C
 - Find the probability that $0 \leq x \leq 1$



HAMDARD UNIVERSITY BANGLADESH



Fac
Dep

ch: 25th
Date: 17/03/2025

- Q5. (a) Find the expected value of the number of points that will be obtained in a single toss of a fair dice. CLO3 [2]
- (b) Find the characteristic function of $f(x) = e^{-x}, 0 \leq x \leq \infty$ and hence find mean and variance of $f(x)$ CLO3 [5]
- (b) Two unbiased dice are tossed simultaneously. What is the probability of getting a total of point 10 or even numbers from both the dice? CLO3 [3]

- Q6. (a) Seven coins are tossed at a time and the number of head are noted. The experiment is repeated 130 times and the distribution is obtained as follows: CLO3 [6]

No of heads	0	1	2	3	4	5	6	7
Frequencies	7	6	19	35	30	23	7	1

Fit a binomial distribution to the above data assuming

- The coin is unbiased
 - The nature of the coin is not known.
- (b) A Random variable x has the following probability density function, CLO3 [4]
- $$f(x) = cx(2 - x), 0 \leq x \leq 2$$
- Determine C
 - Find the probability that $0 \leq x \leq 1$

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE 215: Data Structures

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

atch: 25th

Date: 17/03/2025

Final Examination

5×8= 40

Answer all the five questions from the following:

CLOs Marks

CLO1 4

CLO1 4

1. (a) Explain the distinctions between full binary tree and complete binary tree.
(b) Define a priority queue. Construct a priority queue using the given dataset.

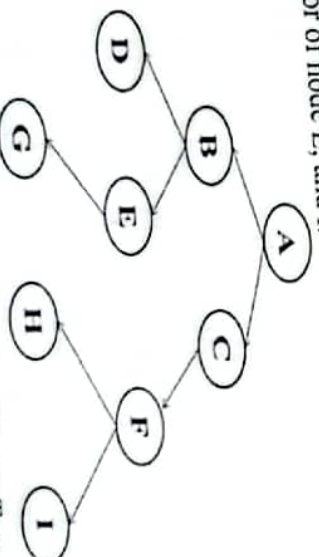
Info	Priority	Link
250	3	6
450	2	5
350	4	
150	1	1
550	3	0
650	2	4
750	4	2

CLO2 2

CLO1 6×1=6

2. (a) State the algorithm used for searching a node in a Binary Search Tree (BST).
(b) Identify the followings from the given binary tree:

- i. Root node.
- ii. Internal nodes.
- iii. Depth.
- iv. Leaf nodes.
- v. Antecedent of node G, I, and F.
- vi. Ancestor of node E, and I.



CLO4 4

3. (a) Assess the efficiency of a Binary Tree, Complete Binary Tree, and Binary Search Tree in terms of time complexity.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE 215: Data Structures

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

atch: 25th

Date: 17/03/2025

Final Examination

5×8= 40

Answer all the five questions from the following:

CLOs Marks

CLO1 4

CLO1 4

1. (a) Explain the distinctions between full binary tree and complete binary tree.
(b) Define a priority queue. Construct a priority queue using the given dataset.

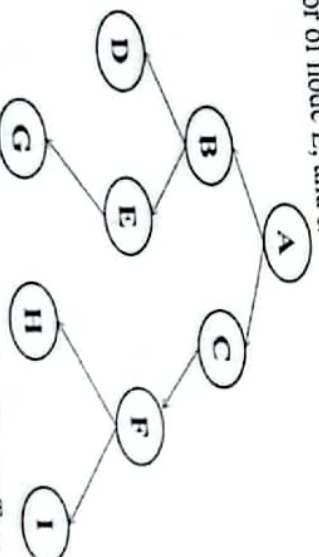
Info	Priority	Link
250	3	6
450	2	5
350	4	
150	1	1
550	3	0
650	2	4
750	4	2

CLO2 2

CLO1 6×1=6

2. (a) State the algorithm used for searching a node in a Binary Search Tree (BST).
(b) Identify the followings from the given binary tree:

- i. Root node.
- ii. Internal nodes.
- iii. Depth.
- iv. Leaf nodes.
- v. Antecedent of node G, I, and F.
- vi. Ancestor of node E, and I.

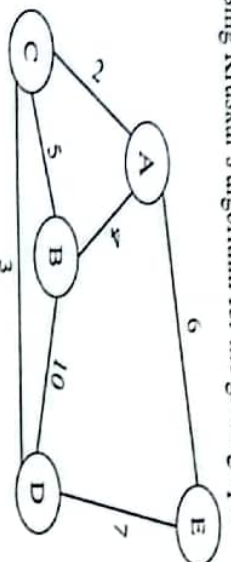


CLO4 4

3. (a) Assess the efficiency of a Binary Tree, Complete Binary Tree, and Binary Search Tree in terms of time complexity.

(b) Design a MST using Kruskal's algorithm for the given graph.

CLO1



4. (a) State the properties of Minimum Spanning Tree (MST).

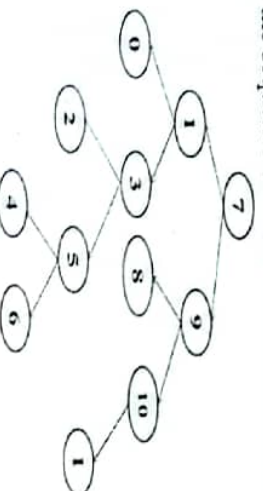
CLO1

2

(b) Apply pre-order, in-order, and post-order traversal techniques to the given binary tree and demonstrate the sequence of nodes visited in each case.

CLO2

6



5. (a) Demonstrate the insertion of elements 30, 35, 28, 24, 32, 19, and 26 into a max heap and explain how the heap property is maintained at each step.

CLO2

4

(b) Perform Bucket Sort on the following list: 29, 34, 47, 15, 7, 88, 92, 61, 5, 75 and show the intermediate steps, assuming each bucket can hold a range of 10 values.

CLO2

4

6. (a) Describe the algorithmic approach of ternary search.

CLO2

4

(b) List the key differences between binary search and ternary search.

CLO4

4

7. (a) Compare the time complexities of binary tree, complete binary tree, and binary search tree for various operations.

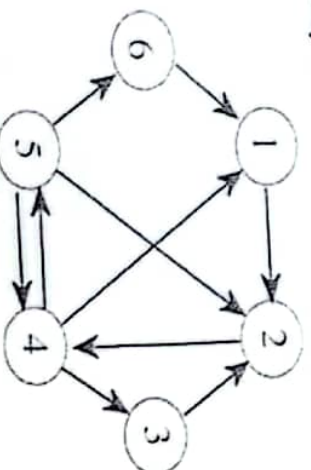
CLO2

4

(b) Construct the adjacency matrix and list for the following graph:

CLO1

4

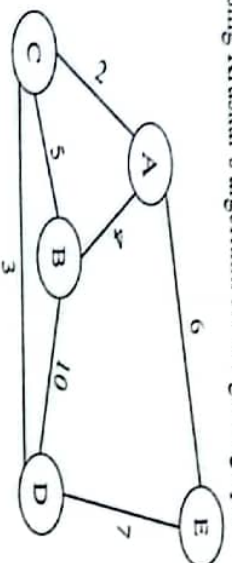


FACULTY
ACT

19/03/2025

(b) Design a MST using Kruskal's algorithm for the given graph.

CLO1



4. (a) State the properties of Minimum Spanning Tree (MST).

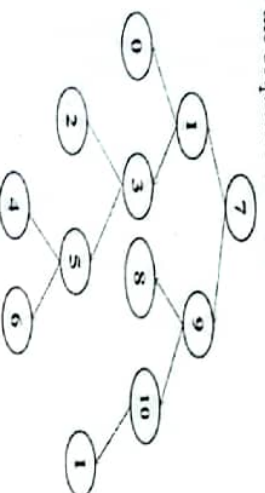
CLO1

2

(b) Apply pre-order, in-order, and post-order traversal techniques to the given binary tree and demonstrate the sequence of nodes visited in each case.

CLO2

6



5. (a) Demonstrate the insertion of elements 30, 35, 28, 24, 32, 19, and 26 into a max heap and explain how the heap property is maintained at each step.

CLO2

4

(b) Perform Bucket Sort on the following list: 29, 34, 47, 15, 7, 88, 92, 61, 5, 75 and show the intermediate steps, assuming each bucket can hold a range of 10 values.

CLO2

4

6. (a) Describe the algorithmic approach of ternary search.

CLO2

4

(b) List the key differences between binary search and ternary search.

CLO4

4

7. (a) Compare the time complexities of binary tree, complete binary tree, and binary search tree for various operations.

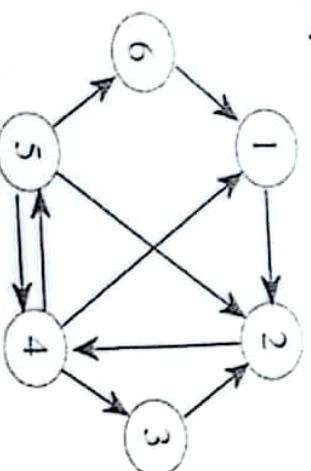
CLO2

4

(b) Construct the adjacency matrix and list for the following graph:

CLO1

4



FACULTY
ACT

19/03/2025



FACULTY OF BUSINESS ADMINISTRATION

ACT215: Financial and Managerial Accounting

Semester: Fall 2024

19/03/2025

Final Examination 2025

Time: 120 minutes

Total Marks: 10x4=40

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.
- Answer any **Four (4)** of the following questions.

- 1 James Smith owns and manages a computer repair service, which had the following trial balance on September 30, 2024. 10

COMPUTER REPAIR SERVICE

Trial Balance

September 30, 2024

	Dr. (Tk.)	Cr. (Tk.)
Cash	12,000	
Accounts Receivable	18,000	
Parts Inventory	15,000	
Prepaid rent	4,000	
Shop Equipment	22,000	
Accounts Payable		23,000
Jack ShellenKamp, Capital		48,000
	71,000	71,000

Summarized transactions for October 2025 were as follows:

- October 1, 2025; Advertising costs, paid in cash, \$1,000.
" 5; Additional repair parts inventory acquired on account \$4,000.
" 8; Miscellaneous expenses, paid in cash, \$2,000.
" 12; Cash collected from customers in payment of accounts receivable \$14,000.
" 20; Cash paid to creditors for accounts payable due \$15,000.
" 25; Repair parts used during January Tk. 4,000
" 27; Repair services performed during January: for cash \$6,000; on account \$9,000.
" 28; Wages for January, paid in cash, \$3,000.
" 30; Jack's drawings during January were \$3,000.

List the journal entries to the accounts in the ledger.

- 2 Explain the following terms:

- a) Contribution Margin (CM);
- b) Break even analysis;
- c) Assumption of CVP;
- d) Intangibles assets;
- e) Adjusting entries.

CLO3 (Analyzing)

02 02 02 02 02

CLO5 (Understanding)



FACULTY OF BUSINESS ADMINISTRATION

ACT215: Financial and Managerial Accounting

Semester: Fall 2024

19/03/2025

Final Examination 2025

Time: 120 minutes

Total Marks: 10x4=40

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.
- Answer any **Four (4)** of the following questions.

- 1 James Smith owns and manages a computer repair service, which had the following trial balance on September 30, 2024. 10

COMPUTER REPAIR SERVICE

Trial Balance

September 30, 2024

	Dr. (Tk.)	Cr. (Tk.)
Cash	12,000	
Accounts Receivable	18,000	
Parts Inventory	15,000	
Prepaid rent	4,000	
Shop Equipment	22,000	
Accounts Payable		23,000
Jack ShellenKamp, Capital		48,000
	71,000	71,000

Summarized transactions for October 2025 were as follows:

- October 1, 2025; Advertising costs, paid in cash, \$1,000.
" 5; Additional repair parts inventory acquired on account \$4,000.
" 8; Miscellaneous expenses, paid in cash, \$2,000.
" 12; Cash collected from customers in payment of accounts receivable \$14,000.
" 20; Cash paid to creditors for accounts payable due \$15,000.
" 25; Repair parts used during January Tk. 4,000
" 27; Repair services performed during January: for cash \$6,000; on account \$9,000.
" 28; Wages for January, paid in cash, \$3,000.
" 30; Jack's drawings during January were \$3,000.

List the journal entries to the accounts in the ledger.

- 2 Explain the following terms:

- a) Contribution Margin (CM);
- b) Break even analysis;
- c) Assumption of CVP;
- d) Intangibles assets;
- e) Adjusting entries.

CLO3 (Analyzing)

CLO5 (Understanding)

02 02 02 02 02



3. Monti River Resort opened for business on June 1 with eight air-conditioned units. Its trial balance before adjustment on August 31 is as follows.

MONI RIVER RESORT
Trial Balance
August 31, 2024

	TK.	TK.
Tiles of Accounts	39,600	
Cash	2,300	
Supplies	5,000	
Prepaid Insurance	22,000	
Land	1,20,000	
Cottages	26,000	
Furniture		6,000
Accounts Payable		80,000
Unearned Rent	8,400	
Mortgage payable	70,000	
Neosho's Capital	1,20,000	
Neosho's Drawing		6,000
Rent Revenue		3,600
Repair Expense		51,000
Salaries Expense		9,400
Utilities Expense		2,84,900
		<u>2,84,900</u>

Other data:

- Insurance expires at the rate of TK.900 per month.
- A count of supplies shows TK.950 of unused supplies on August 31.
- Unearned rent of TK.3,900 was earned.
- Salaries TK.800 are accrued and unpaid at August 31.
- The Mortgage interest rate is 10%. (The Mortgage was taken out on August 1).

Requirement:

- Make up adjusting entries at the end of the month of August 31, 2024.
- Create adjusted trial Balance on August 31, 2024.

4. Jarir Corporation manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below:

Particular	Total	Per unit	% of sales
Sales (21,000 units)	12,60,000	TK.60	100%
Variable expenses	9,45,000	TK.45	7%
Contribution margin	3,15,000	TK.15	2%
Fixed expenses	2,40,000		
Net operating income	75,000		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.



3. Monti River Resort opened for business on June 1 with eight air-conditioned units. Its trial balance before adjustment on August 31 is as follows.

MONI RIVER RESORT
Trial Balance
August 31, 2024

	TK.	TK.
Tiles of Accounts	39,600	
Cash	2,300	
Supplies	5,000	
Prepaid Insurance	22,000	
Land	1,20,000	
Cottages	26,000	
Furniture		6,000
Accounts Payable		80,000
Unearned Rent	8,400	
Mortgage payable	70,000	
Neosho's Capital	1,20,000	
Neosho's Drawing		6,000
Rent Revenue		3,600
Repair Expense		51,000
Salaries Expense		9,400
Utilities Expense		2,84,900
		<u>2,84,900</u>

Other data:

- Insurance expires at the rate of TK.900 per month.
- A count of supplies shows TK.950 of unused supplies on August 31.
- Unearned rent of TK.3,900 was earned.
- Salaries TK.800 are accrued and unpaid at August 31.
- The Mortgage interest rate is 10%. (The Mortgage was taken out on August 1).

Requirement:

- Make up adjusting entries at the end of the month of August 31, 2024.
- Create adjusted trial Balance on August 31, 2024.

4. Jarit Corporation manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below:

Particular	Total	Per unit	% of sales
Sales (21,000 units)	12,60,000	TK.60	100%
Variable expenses	9,45,000	TK.45	7%
Contribution margin	3,15,000	TK.15	2%
Fixed expenses	2,40,000		
Net operating income	75,000		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

out:

10

The company's CM ratio and variable expenses ratio.

The company's break-even point in both sales revenue and units.

iii) Assume that sales increased by Tk.4,50,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase?

iv) As per the original data, Assume that next year management wants the company to earn a profit of at least Tk. 1,20,000. How many units will have to be sold to meet this target profit?

v) Refers to the original data, compute the company's margin of safety in both amount and percentage.

5

04
06

- Define the different errors in trial balance. Discuss the purpose of trial balance.
- The following Trial Balance of Mr. James Gomez has been prepared incorrectly. You are asked to Construct the Trial Balance correctly:-

December 31, 2024

Trial Balance

Account titles	Debit(TK)	Credit (Tk)
Capital – Mr. Gomez	3,90,000	22,480
Drawings- Mr. Gomez		32,000
Furniture	2,000	
Import Duty	3,000	
Export Duty	18,720	
Wages	2,000	
Allowance for doubtful debt	840	
Bad Debt	240	
Insurance Premium	50,000	
Buildings		32,900
Transport in	5,000	
Transportation out	2,500	
Salaries		3,07,700
Sales		1,42,230
Purchases	42,790	
Accounts Payable	1,500	
Notes Receivable	28,460	
Notes payable		11,760
Sales Return		15,000
Purchase Return		37,800
Accounts Receivable		25,000
Investment	50,000	
Cash	12,270	
Bank Loan		4,000
Advertisement		46,420
Inventory		

CLO3 (Creating)

out:

10

The company's CM ratio and variable expenses ratio.

The company's break-even point in both sales revenue and units.

iii) Assume that sales increased by Tk.4,50,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase?

iv) As per the original data, Assume that next year management wants the company to earn a profit of at least Tk. 1,20,000. How many units will have to be sold to meet this target profit?

v) Refers to the original data, compute the company's margin of safety in both amount and percentage.

5

04
06

- Define the different errors in trial balance. Discuss the purpose of trial balance.
- The following Trial Balance of Mr. James Gomez has been prepared incorrectly. You are asked to Construct the Trial Balance correctly:-

December 31, 2024
Trial Balance

Account titles	Debit(TK)	Credit (Tk)
Capital – Mr. Gomez	3,90,000	22,480
Drawings- Mr. Gomez		32,000
Furniture	2,000	
Import Duty	3,000	
Export Duty	18,720	
Wages	2,000	
Allowance for doubtful debt	840	
Bad Debt	240	
Insurance Premium	50,000	
Buildings		32,900
Transport in	5,000	
Transportation out	2,500	
Salaries		3,07,700
Sales		1,42,230
Purchases	42,790	
Accounts Payable	1,500	
Notes Receivable	28,460	
Notes payable		11,760
Sales Return		15,000
Purchase Return		37,800
Accounts Receivable		25,000
Investment	50,000	
Cash	12,270	
Bank Loan		4,000
Advertisement		46,420
Inventory		

CLO3 (Creating)

out:

10

The company's CM ratio and variable expenses ratio.

The company's break-even point in both sales revenue and units.

iii) Assume that sales increased by Tk.4,50,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase?

iv) As per the original data, Assume that next year management wants the company to earn a profit of at least Tk. 1,20,000. How many units will have to be sold to meet this target profit?

v) Refers to the original data, compute the company's margin of safety in both amount and percentage.

5

04
06

- Define the different errors in trial balance. Discuss the purpose of trial balance.
- The following Trial Balance of Mr. James Gomez has been prepared incorrectly. You are asked to Construct the Trial Balance correctly:-

December 31, 2024
Trial Balance

Account titles	Debit(TK)	Credit (Tk)
Capital – Mr. Gomez	3,90,000	22,480
Drawings- Mr. Gomez		32,000
Furniture	2,000	
Import Duty	3,000	
Export Duty	18,720	
Wages	2,000	
Allowance for doubtful debt	840	
Bad Debt	240	
Insurance Premium	50,000	
Buildings		32,900
Transport in	5,000	
Transportation out	2,500	
Salaries		3,07,700
Sales		1,42,230
Purchases	42,790	
Accounts Payable	1,500	
Notes Receivable	28,460	
Notes payable		11,760
Sales Return		15,000
Purchase Return		37,800
Accounts Receivable		25,000
Investment	50,000	
Cash	12,270	
Bank Loan		4,000
Advertisement		46,420
Inventory		

CLO3 (Creating)



HAMDARD UNIVERSITY BANGLADESH

- 6 George began operations as a private investigator on January 1, 2024. The trial balance columns of the worksheet for Thomas Magnum, P.I. at March 31 are as follows.

GEORGE, P.I.

For the Quarter Ended March 31, 2024

Trial Balance

Account titles	Dr. (Tk)	Cr. (Tk.)
Cash	11,400	
Accounts Receivable	6,620	
Supplies	1,050	
Prepaid Insurance	2,400	
Equipment	30,000	
Notes Payable		11,000
Accounts Payable		12,350
T. Magnum, Capital		20,000
T. Magnum, Drawing	1,600	14,620
Service Revenue		
Salaries Expense	2,200	
Travel Expense	1,300	
Rent Expense	1,200	
Miscellaneous Expense	200	
Total	57,970	57,970

Other data:

- Supplies on hand total Tk.480.
- Depreciation is Tk.1,100 per quarter.
- Interest accrued on 6-month note payable, issued January 1, Tk.500.
- Insurance expires at the rate of TK.350 per month.
- Services provided but unbilled at March 31 total Tk.580.

You are required to

- Construct a complete worksheet with ten columns.
- Illustrate journalize adjusting entries from the adjustments columns of the worksheet.

TAKDHUM Scale



HAMDARD UNIVERSITY BANGLADESH

- 6 George began operations as a private investigator on January 1, 2024. The trial balance columns of the worksheet for Thomas Magnum, P.I. at March 31 are as follows.

GEORGE, P.I.

For the Quarter Ended March 31, 2024

Trial Balance

Account titles	Dr. (Tk)	Cr. (Tk.)
Cash	11,400	
Accounts Receivable	6,620	
Supplies	1,050	
Prepaid Insurance	2,400	
Equipment	30,000	
Notes Payable		11,000
Accounts Payable		12,350
T. Magnum, Capital		20,000
T. Magnum, Drawing	1,600	14,620
Service Revenue		
Salaries Expense	2,200	
Travel Expense	1,300	
Rent Expense	1,200	
Miscellaneous Expense	200	
Total	57,970	57,970

Other data:

- Supplies on hand total Tk.480.
- Depreciation is Tk.1,100 per quarter.
- Interest accrued on 6-month note payable, issued January 1, Tk.500.
- Insurance expires at the rate of TK.350 per month.
- Services provided but unbilled at March 31 total Tk.580.

You are required to

- Construct a complete worksheet with ten columns.
- Illustrate journalize adjusting entries from the adjustments columns of the worksheet.

TAKDHUM Scale

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 213: Digital Electronics

Batch: 25th
Date: 15/03/2025

Final Examination

Semester: Fall 2024
Time: 2 Hours
Total Marks: 40

Answer any five questions:

5 × 8 = 40

- | | CLOs | Marks |
|--|------|-----------|
| 1. (a) Illustrate functional diagram of an Arithmetic-Logic Unit. | CLO3 | 4 |
| (b) Design a five-bit parallel adder circuit using full adders. | CLO1 | 4 |
| 2. (a) Sketch a 16-bit synchronous Up/Down counter. | CLO4 | 4 |
| (b) Draw a MOD-25 synchronous Up counter using D flip-flop. | CLO4 | 4 |
| 3. (a) Write the differences between parallel in/parallel out and parallel in/serial out register. | CLO4 | 4 |
| (b) List the commonly used digital IC terminology. | CLO2 | |
| 4. (a) Illustrate a Binary Coded Decimal (BCD) Decoder. | CLO1 | 4 |
| (b) Draw and explain an eight input Encoder. | CLO1 | 4 |
| 5. (a) Write the steps to build a small project management. | CLO1 | 4 |
| (b) List the commonly used IC packages. | CLO2 | 4 |
| 6. (a) Which memory stores the most bits: a 5M × 16 memory or a memory that stores 3M words at a word size of 32 bits? | CLO4 | 4 |
| (b) Draw and explain a Digital-to-Analog Converter. | CLO1 | 4 |
| 7. Write short notes on: | CLO4 | 2 × 4 = 8 |
| I. Static RAM | | |
| II. Access time | | |
| III. Auxiliary memory | | |
| IV. Capacity | | |